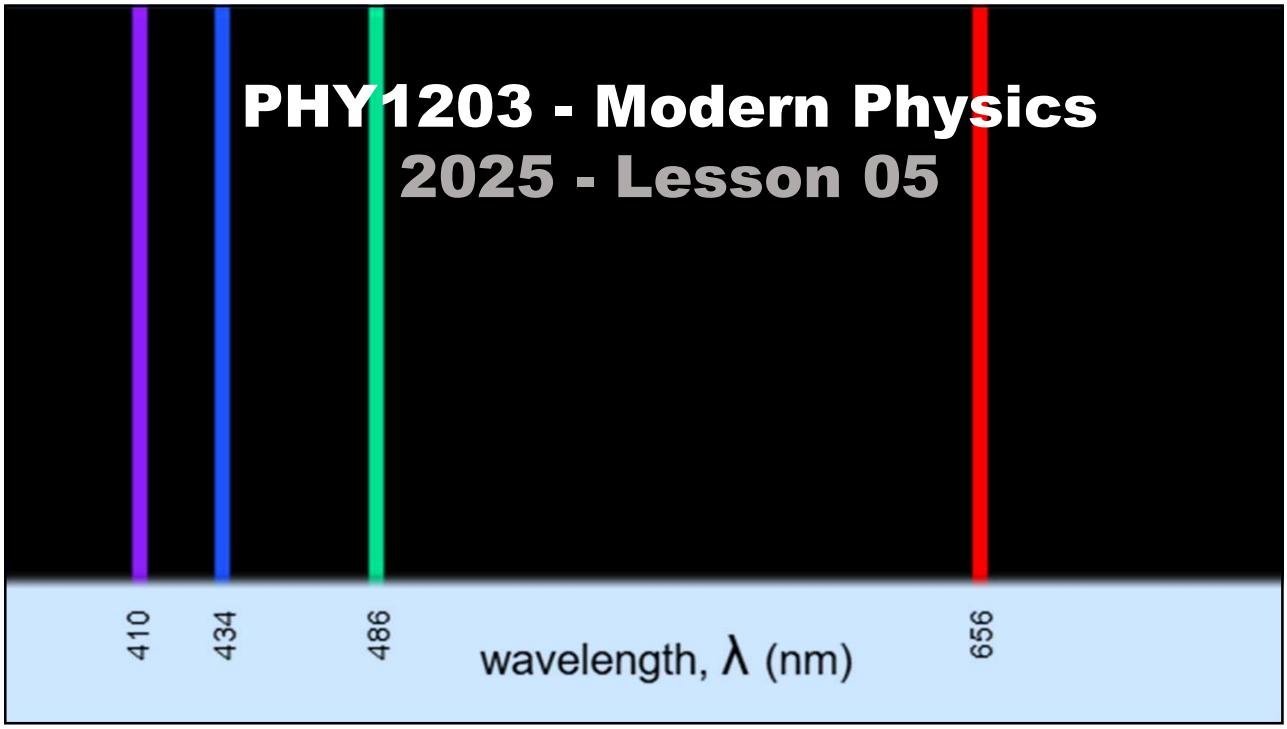


Bohr model

Niels Bohr proposed the Bohr atomic model in 1913.

He described the atom as a small, positively charged nucleus surrounded by electrons in specific, quantized orbits.

These orbits are at fixed distances from the nucleus, and electrons can only move in these defined paths, not in between.

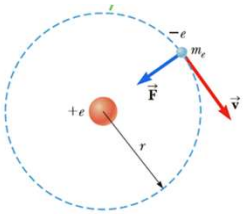


Niels Bohr (1885-1962)

Bohr model

1. Quantized Energy Levels

Electrons move in well defined circular orbits about the positive nucleus under the influence of the Coulomb force of attraction.



Bohr model of the hydrogen atom

2. No Radiation in Orbits

Only certain electron orbits are stable and allowed . Electrons moving in these orbits do not emit electro-magnetic radiation and therefore, the total energy of the atom remains constant.

3. Quantum Jumps

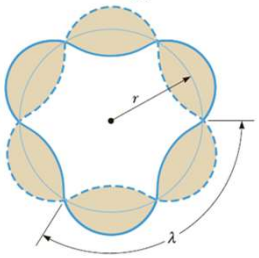
Radiation is emitted by the H atom when the electron "jumps" from a higher energy levels (away from the nucleus) to a lower energy level. Similarly by absorbing energy it can jump to a to higher energy level . The frequency f of the radiation is given by

$$E_i - E_f = hf$$

4. Angular Momentum Quantization

The circumference of an orbit must contain an integral number of de Broglie wavelength, $2\pi r = n\lambda$ where $n = 1, 2, 3, \ldots \ldots \ldots$.

The de Broglie wavelength of an electron is given by $\lambda = \frac{h}{m_e v}$.



Standing-wave pattern for an electron in a stable orbit of hydrogen. There are 3 full wavelength in this orbit.

$$2\pi r = n\frac{h}{m_e v}$$

$$m_e v r = n\frac{h}{2\pi}$$

$$pr = n\hbar \quad \text{where } n = 1, 2, 3, \ldots \ldots \ldots.$$

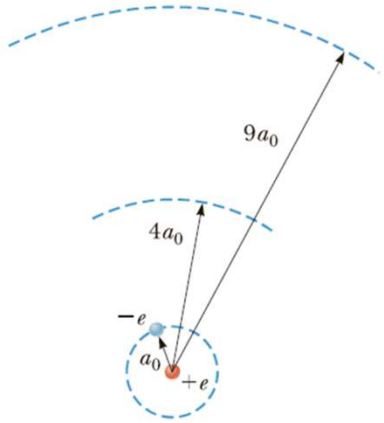
Therefore, the angular momentum of an electron in an orbit is quantized, meaning it can only take on specific discrete values.

Explanation of the Atomic Spectra

The Bohr model was successful in explaining the line spectrum of hydrogen. When an electron transitions between energy levels, it emits or absorbs a photon of a specific energy and frequency, corresponding to a specific line in the spectrum.

The Bohr model has limitations because it only works for hydrogen but fails to accurately predict the spectra of atoms with many electrons.

Bohr model of hydrogen atom



The first three circular orbits predicted by the Bohr model of hydrogen atom.

$$r_n = \frac{n^2\hbar^2}{m_e k_e e^2} \qquad n = 1, 2, 3, \ldots$$

where k_e is the kinetic energy of electron

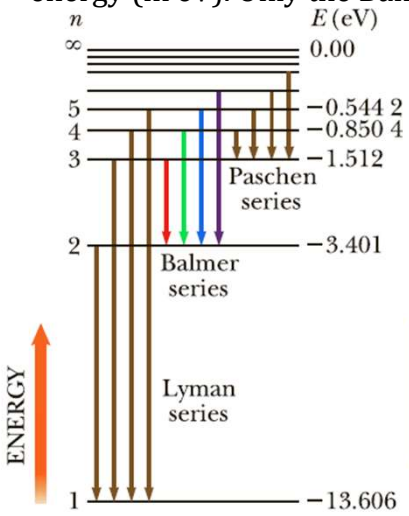
The orbit with the smallest radius is called the **Bohr radius** a_0 corresponding to n =1 and has the value

$$a_0 = \frac{\hbar^2}{m_e k_e e^2}$$

= 0.0529 nm

Bohr Model - Hydrogen Spectra

The energy level diagram for hydrogen. with quantum numbers and energy (in eV). Only the Balmer series has visible light.



$$E_n = -\frac{m_e k_e^2 e^4}{2\hbar^2}\left(\frac{1}{n^2}\right) \qquad n = 1, 2, 3, \ldots$$

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

$$f = \frac{E_i - E_f}{\hbar} = \frac{m_e k_e^2 e^4}{4\pi\hbar^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \qquad n_f < n_i$$

$$E_n = -\frac{m_e k_e^2 e^4}{2\hbar^2}\left(\frac{1}{n^2}\right) \qquad n = 1, 2, 3, \ldots$$

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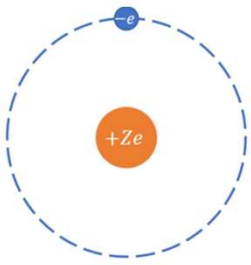
$$\frac{1}{\lambda} = R_{\text{H}}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \qquad \text{where } R_{\text{H}} = \frac{m_e k_e^2 e^4}{4\pi c\hbar^3}$$

Hydrogen like atoms

The Bohr theory can be generalized to hydrogen-like atoms, such as singly ionized helium or doubly ionized lithium. This modification consists of replacing e^2 by Ze^2 wherever it occurs.

$$E_n = -\frac{m_e k_e^2 Z^2 e^4}{2\hbar^2}\left(\frac{1}{n^2}\right) \qquad n = 1, 2, 3, \ldots$$

$$\frac{1}{\lambda} = \frac{m_e k_e^2 Z^2 e^4}{4\pi c\hbar^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$



The failure of the Bohr model

Reasons for the failure:

(1) Multi-Electron Atoms:

The model was successful only for hydrogen-like species (single-electron atoms or ions) and could not explain the complex spectra of atoms with more than one electron.

(2) Heisenberg Uncertainty Principle:

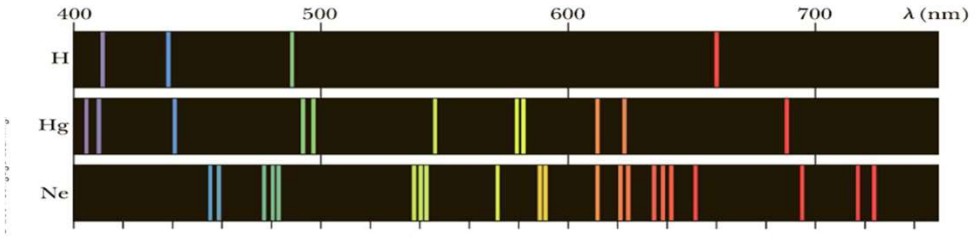
The model assumes electrons have precise orbits, meaning both their position and momentum are known simultaneously. This contradicts the Heisenberg uncertainty principle.

(3) Inability to Explain Spectral Effects:

(a) Fine Structure: Bohr's model couldn't explain the splitting of spectral lines into finer lines.

(b) Zeeman Effect: It failed to explain the splitting of spectral lines in the presence of an external magnetic field.

(c) Stark Effect: It also couldn't explain the splitting of spectral lines when atoms are subjected to an electric field.



(4) Dual Nature of Electrons:

The Bohr model assumes that electron as particles in fixed orbits and does not consider their wave-like nature of de Broglie's hypothesis.

(5) Molecular Bonding:

The model did not provide any support for understanding how atoms bond to form molecules.

(6) Atomic Geometry:

The Bohr model suggests a flat, circular structure for atoms, which is incorrect; atoms have a more spherical, three-dimensional shape.

X-rays

- While experimenting on cathode rays, Wilhelm Röntgen accidentally discovered X-rays in 1895.
- Cathode rays are a fast-moving electron beam. When high-energy electrons collide with a hard metal and are instantly decelerated, part of their kinetic energy is released as X-rays.
- X-rays are electromagnetic radiations with very short wavelengths (0.01 nm –10 nm). It is an ionizing radiation.

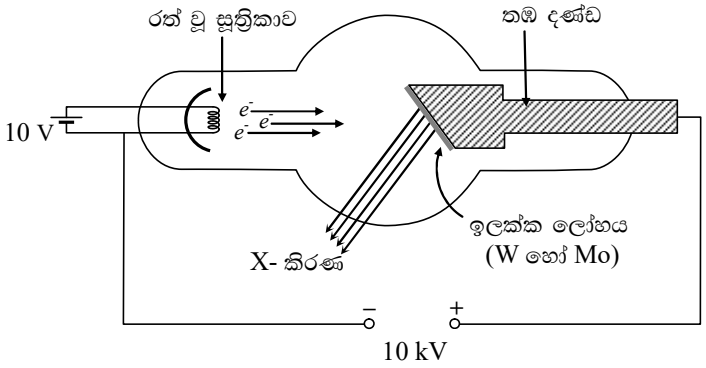
The old X-ray tube.

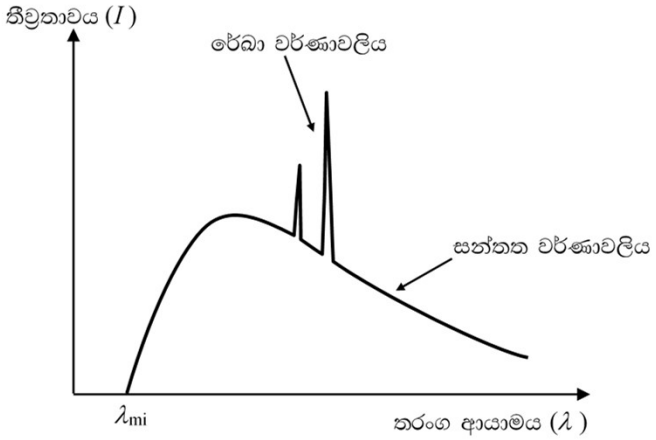


X-ray production

First electrons are produced by thermionic emission of a heated filament and accelerated using a very high potential difference (10 kV).

When these high-energy electrons collide with a hard target metal with a high boiling point (e.g. W or Mo), X-rays are emitted.





$$hf_{\max} = eV$$

$$h\frac{c}{\lambda_{\min}} = eV$$

$$\lambda_{\min} = \frac{hc}{eV}$$

A continuous X-ray spectrum (Bremsstrahlung Radiation) and sharp spikes with high intensity (Characteristic Radiation) are obtained.

The spectrum has a minimum wavelength (λ_{min}) corresponding to the maximum energy of the electrons.
The value of λ_{min} depends on the high potential difference (tube voltage) applied to accelerate the electrons.

The intensity of the beam increases with increasing filament current.
The line spectrum is a characteristic of the target metal used.

Depending on the tube voltage applied between the anode and cathode of the tube, the electrons are accelerated and collide with the target metal with a high velocity (kinetic energy).

The remaining electron energy is converted into heat. A copper rod is used to conduct that heat energy out of the X-ray tube, and a copper tube carrying water at the end of the copper rod cools it.

The thermionic electrons released from the hot filament are accelerated due to the high potential difference between the anode and the cathode and hit the target metal at a high velocity (or energy).

Kinetic energy of an electron when hitting $K_{\mathrm{m}} = eV$ potential energy gained

A small part (<2%) of this is converted into X-ray photon which is an e-m radiation.

The maximum energy of a photon is the maximum energy of an electron just before hitting the target. There is a maximum frequency or minimum wavelength (λ_{min}) for that photon. $h\frac{c}{\lambda_{\mathrm{min}}} = eV$

$$\lambda_{\mathrm{min}} = \frac{hc}{eV}$$

Starting from (λ_{min}) a continuously increasing wavelengths are produced in the X-ray tube.

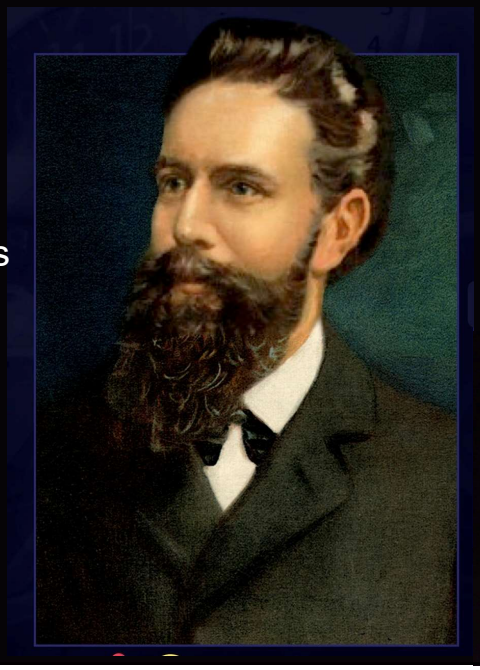
The first X-ray of a human was Anna Bertha Ludwig's hand on December 22, 1895, taken by her husband, Röntgen.

It showed bones and the ring and she has shouted "I have seen my death".

This image is named ***Hand mit Ringen***.



The first Nobel Prize for Physics in 1901 was awarded to Wilhelm Röntgen for his discovery of X-rays





Properties of X-rays

(1) Since it is an electromagnetic wave, it travels at the speed of light in a vacuum.

(2) It is not deflected by electric or magnetic fields.

(3) It has the ability to penetrate solids. As the density of the medium increases, the penetration distance decreases. X-rays can be stopped by a lead plate about 1 mm thick.

(4) Since it penetrates through matter it cannot be focused by lenses or reflected by mirrors.

(5) When the wavelength of X-rays is close to the spacing between atomic layers in a crystal lattice, diffraction (XRD) patterns are produced.

(6) Can ionize gases and molecules.

(7) Photographic plats are developed when exposed to X-rays

(8) Causes fluorescence when incident on certain minerals.

(9) Causes photoelectric emission.

e.g. (1) Determine the minimum wavelength of X-rays produced by a tube voltage of 10 kV.

$e = 1.6 \times 10^{-19}$ C $\quad h = 6.6 \times 10^{-34}$ J s
$V = 10^{4}$ V $\quad c = 3 \times 10^{8}$ m s^{-1}

$$\lambda_{\min} = \frac{hc}{eV}$$

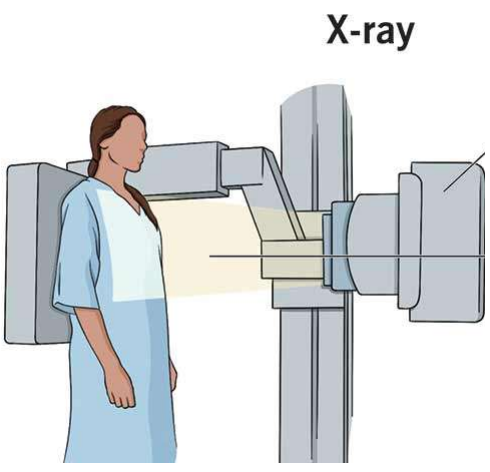
$$\lambda_{\min} = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{1.6 \times 10^{-19} \times 10^{4}}$$

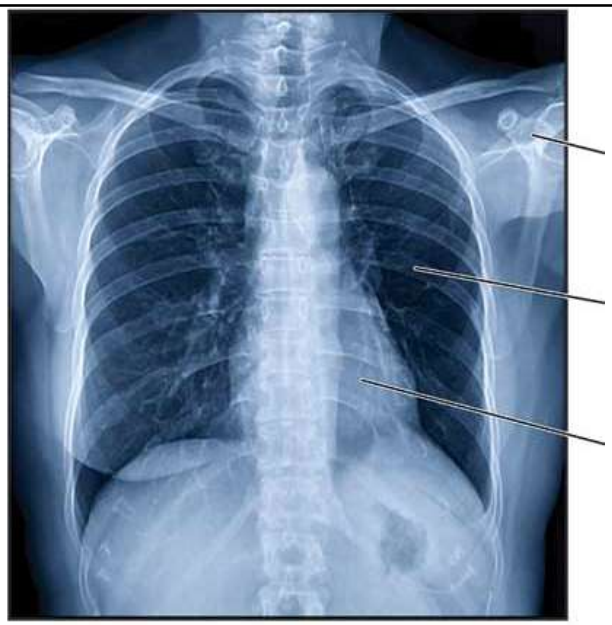
$$\lambda_{\min} = 1.23 \times 10^{-10} \text{ m}$$

$$= 1.23 \text{ Å}$$

1. In medicine

To take an X-ray image inside the body

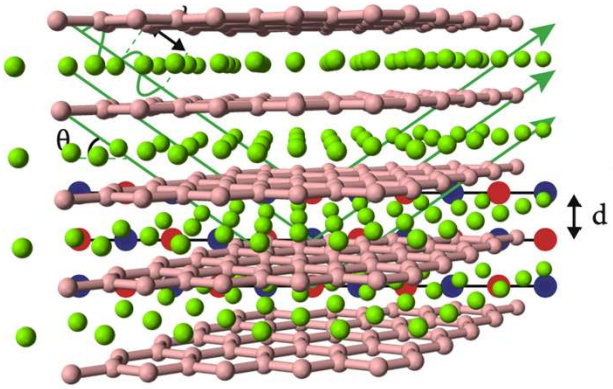




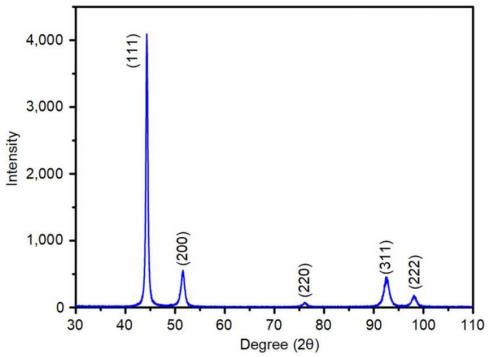
X- ray image

2. In scientific research

To investigate the lattice structure of crystalline compounds.

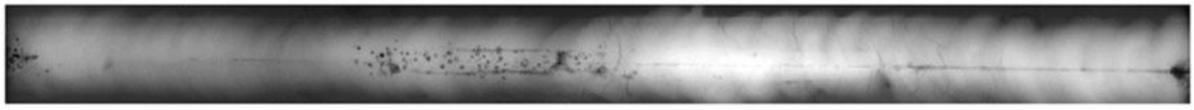


X-ray diffraction (XRD) pattern

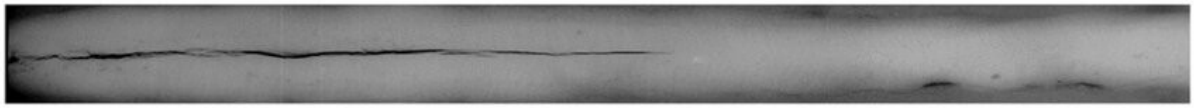


3. In engineering applications

To find the defects and flaws inside metal part in machinery.



(a)



(b)

4. In security checkups

In airports to rapidly check goods inside bag gages without opening

